Shubham Singh

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Bio

I am an **AI and ML researcher** with a focus on **multi-objective optimization** for sociotechnical systems. My Ph.D. thesis identifies the causes of suboptimal performance of ML algorithms evaluated on efficiency and fairness trade-offs. I develop empirical and theoretical methods that provide the Pareto-optimal solutions and apply them to resource allocation, ranking, and prediction tasks. I have 10 years of professional experience developing ML algorithms, mathematical models, and web applications. My research has been published at major AI and security conferences and workshops like EAAMO, USENIX, ICML, and NeurIPS, with **280+ citations**.

Research Interests: Responsible Machine Learning, AI Evaluations, Computational Economics, Privacy & Security

Education

University of Illinois Chicago — *Aug 2019–Dec 2025*
Ph.D. in Computer Science

- **Thesis:** "Fair Scheduling and Resource Allocation for Public Services"
- **Committee:** Chris Kanich, Ian A. Kash, Mesrob I. Ohannessian, Elissa M. Redmiles, Lu Cheng

University of Illinois Chicago — *Aug 2022–May 2024*
M.S. in Computer Science

- **Coursework:** Advanced Algorithms, Advanced Machine Learning, Mathematical Theory of Artificial Intelligence, Computational Finance, Causal Inference and Learning.

Indraprastha Institute of Information Technology Delhi — *Aug 2012–May 2016*
B.Tech. in Computer Science

- **Coursework:** Machine Learning, Designing Human Centered Systems, Mobile Computing, Image Analysis

Experience

Research Scientist (Postdoctoral Researcher) — *Chicago, IL*
University of Chicago Data Science Institute — *Nov 2025–Present*

- **Social Choice Theory.** Create a bridge between applied ML for social good and computational social science.
- Advisor: Moon Duchin

Research Assistant — *Chicago, IL*
University of Illinois Chicago — *Aug 2019–Nov 2025*

- **Machine Learning.** Showed theoretical limitations of scoring-based ranking methods for efficiency and fairness trade-offs, and suggested approximate algorithms that dominate the state-of-the-art by 20%. Improved the machine learning-based food inspection schedule used by the Chicago Department of Public Health (CDPH) by increasing efficiency by 11% and fairness by 76%.
- **Resource Allocation.** Studied the effect of resources on efficiency and fairness for allocation problems using homogeneous functions, characterizing the nuanced conclusions drawn from a seemingly straightforward choice of objectives, and showing the strong upper and lower bounds for the gap in the best and worst outcomes. In addition, performed empirical analyses on the redistricting data for Chicago to study the effect of voter turnout and changes in city wards, informing resource allocation to each precinct.
- **LLM Evaluation.** In collaboration with industry and academic researchers, conducted an extensive review of evaluation methods for LLMs, highlighting the shortcomings and recommending improved methods to evaluate the societal, economic, cultural, and environmental impacts of Large Language Models (LLMs). With another group of interdisciplinary researchers from CS and Law, we propose the integration of generative AI into the EU's Digital Markets Act (DMA).
- **Privacy & Security.** Designed and conducted an in-lab user study that uses behavioral economics to understand people's valuation of their online accounts, useful for designing effective privacy and security

policies. Collected and studied 900+ QR code in the wild to learn the privacy threats to the users and observed no additional threats. Mentored an undergraduate student to study the effect of the YouTube recommendation system on suggestions for child-focused videos and the relationship between videos and advertisements. Developed ML-based system to help users remove sensitive files from online cloud storage and reduce their risk of breach by 26%, live app: cloudsweeper.app.
- Advisors: Chris Kanich, Ian A. Kash, Mesrob I. Ohannessian

Summer Research Assistant *Chicago, IL*
IDEAL Institute, University of Chicago Booth School of Business *May 2023–Aug 2023*
- **Fair Public Transit.** Surveyed the literature to understand fairness notions for public transit. Worked with the NYC MTA dataset to employ a multinomial choice model and estimate time distance between origin-destination pairs. The estimation helped us determine the efficiency and fairness of the train services.
- Advisors: Ozan Candogan, Ian A. Kash

Research Intern *Saarbrücken, Germany*
Max Planck Institute for Software Systems *May 2022–Aug 2022*
- **Behavioral Economics for Online Accounts Value.** Recruited undergraduate and graduate students as participants from MPI-SWS and Saarland University to conduct an in-lab user study and a Prolific online study. Conducted the study to elicit true value of people's online accounts using behavioral economics methods. Presented the preliminary results from the qualitative data analysis at the Cornell, Maryland, Max Planck Research School (CMMRS) 2022.
- Advisors: Oshrat Ayalon, Elissa M. Redmiles

Research Associate *Delhi, India*
Precog Research Lab, Indraprastha Institute of Information Technology Delhi *Oct 2017–Jul 2019*
- Applied deep learning models to detect unsafe content on video streaming platforms, making them 30% safer for children.
- Created node embedding framework for efficient cross-network representation of user profiles, beating the then state-of-the-art approach by 20%.
- Advisors: Ponnurangam Kumaraguru, Arun Balaji Buduru

Full Stack Developer *Bengaluru, India*
Tapzo (Amazon Acquired Startup) *Jun 2016–Sep 2017*
- Saved hundreds of hours in the development cycle by building a generic cross-category coupons module for the Android application, which was used by 7 other teams. Also, developed food, bill payment, and hotel categories from scratch and launched to 5M users.
- Wrote back-end services to track transaction state changes and user actions for providing better user reinforcements that improved the users' post-transaction experience.
- Mentor: Adarsh Pandey

Application Developer Intern *Gurugram, India*
MobiKwik *May 2015–Aug 2015*
- Migrated Android build environment from Maven to Gradle for better dependency management and faster build times.
- Rewrote the application network layer for multi-thread performance and error handling to support low-bandwidth users.
- Mentor: Atul Goyal

Publications

* indicates equal contributions.

[1] Irene Solaiman, Zeerak Talat, William Agnew, Lama Ahmad, Dylan K. Baker, Su Lin Blodgett, Canyu Chen, III Daumé Hal, Jesse Dodge, Isabella Duan, Ellie Evans, Felix Friedrich, Avijit Ghosh, Usman Gohar, Sara Hooker, Yacine Jernite, Pratyusha Ria Kalluri, Alina Leidinger, Alberto Lusoli, Michelle Lin, Xiuzhu Lin, Sasha Luccioni, Jennifer Mickel, Margaret Mitchell, Jessica Newman, Anaelia Ovalle, Marie-Therese Png, **Shubham Singh**, Andrew Strait, Lukas Struppek, Arjun Subramonian, and Apostol Vassilev. "Evaluating the Social Impact of Generative AI Systems". In: *The Oxford Handbook of the Foundations and Regulation of Generative AI*. Oxford University Press. ISBN: 9780198940272. URL: https://doi.org/10.1093/oxfordhb/9780198940272.013.0025.

[2] **Shubham Singh**, Jackie Hu, Cormac Herley, Elissa Redmiles, Siddharth Suri, and Oshrat Ayalon. *To Bid or Not to Bid: Using Auctions to Understand User Valuation of Digital Accounts*. Preprint. 2025.

[3] **Shubham Singh**, Chris Kanich, and Ian A. Kash. *Effect of Resources on the Efficiency-Fairness Tradeoff for Allocation Problems*. Preprint. 2025.

[4] **Shubham Singh**, Ian A. Kash, and Mesrob I. Ohannessian. *Scoring is Not Enough: Addressing Gaps in Fairness-Utility Tradeoff for Ranking*. Preprint. 2025.

[5] Ayse Gizem Yasar, Andrew Chong, Evan Dong, Thomas Krendl Gilbert, Sarah Hladikova, Roland Maio, Carlos Mougan, Xudong Shen, **Shubham Singh**, Ana-Andreea Stoica, Savannah Thais, and Miri Zilka. "Integration of Generative AI in the Digital Markets Act: Contestability and Fairness from a Cross-Disciplinary Perspective". In: *Regulatable ML Workshop @ NeurIPS 2024*. Dec. 2024. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4769439.

[6] Ayse Gizem Yasar, Andrew Chong, Evan Dong, Thomas Krendl Gilbert, Sarah Hladikova, Roland Maio, Carlos Mougan, Xudong Shen, **Shubham Singh**, Ana-Andreea Stoica, Savannah Thais, and Miri Zilka. "AI and the EU Digital Markets Act: Addressing the Risks of Bigness in Generative AI". In: *GenLaw Workshop @ ICML 2023*. July 2023. URL: https://arxiv.org/abs/2308.02033.

[7] Rayaan Siddiqi*, **Shubham Singh***, Lenore Zuck, and Chris Kanich. "Tracking, But Make It Offline: The Privacy Implications of Scanning QR Codes Found in the World". In: *ConPro Workshop @ S&P 2023*. May 2023. URL: https://conpro23.ieee-security.org/papers/siddiqi-conpro23.pdf.

[8] **Shubham Singh**, Bhuvni Shah, Chris Kanich, and Ian A. Kash. "Fair Decision-Making for Food Inspections". In: *Equity and Access in Algorithms, Mechanisms, and Optimization*. EAAMO 2022. Oct. 2022. ISBN: 9781450394772. DOI: 10.1145/3551624.3555289.

[9] Tanya Berger-Wolf, Allison Howell, Chris Kanich, Ian A. Kash, Moniba Keymanesh, Barbara Kowalcyk, Gina Nicholson Kramer, Andrew Perrault, and **Shubham Singh**. "Open Problems in (Un)fairness of the Retail Food Safety Inspection Process". In: *Responsible Decision Making in Dynamic Environments Workshop @ ICML 2022*. July 2022. URL: https://shubhams.github.io/assets/pdf/openproblems_rdmde_2022.pdf.

[10] Mohammad Taha Khan, Christopher Tran, **Shubham Singh**, Dimitri Vasilkov, Chris Kanich, Blase Ur, and Elena Zheleva. "Helping Users Automatically Find and Manage Sensitive, Expendable Files in Cloud Storage". In: *USENIX Security 2021*. Aug. 2021. URL: https://www.usenix.org/conference/usenixsecurity21/presentation/khan-mohammad.

[11] Rishabh Kaushal, **Shubham Singh**, and Ponnurangam Kumaraguru. "NeXLink: Node Embedding Framework for Cross-Network Linkages Across Social Networks". In: *NetSci-X 2020*. 2020. ISBN: 978-3-030-38965-9. DOI: 10.1007/978-3-030-38965-9_5.

[12] **Shubham Singh**, Rishabh Kaushal, Arun Balaji Buduru, and Ponnurangam Kumaraguru. "KidsGUARD: Fine Grained Approach for Child Unsafe Video Representation and Detection". In: *ACM/SIGAPP Symposium on Applied Computing*. SAC 2019. 2019. ISBN: 9781450359337. DOI: 10.1145/3297280.3297487.

Skills

Languages: Python, R, MATLAB, Java, JavaScript, C, C++.

Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Polars, Flask, NodeJS, Celery, Android

Databases: MLflow, Snowflake, Elasticsearch, MongoDB, MySQL, PostgreSQL, Redis

Tools: Git, Docker, VSCode, Jupyter, Vim, IntelliJ, Qualtrics, MaxQDA, LaTeX

LLM Agents: GitHub Copilot, Cursor

Awards & Honors

ACM Travel Grant: $300 *Sep 2025*
ACM EAAMO 2025

ACM Travel Grant: $750 *Sep 2024*
ACM EAAMO 2024

ACM Travel Grant: $750 *Oct 2023*
ACM EAAMO 2023

Google CS Research Mentorship Program *Feb 2023*
Mentor: Sergei Vassilvitskii, Distinguished Scientist and Senior Research Director

Graduate Student Service Award: Public Recognition & $100 *Jun 2021*
UIC CS Graduate Student Association

Employee of the Month: Public Recognition, Certificate & Gift *Feb 2017*
Tapzo (Amazon acquired Startup)

Best Student Mentor: Public Recognition & Certificate *May 2016*
Indraprastha Institute of Information Technology Delhi

Invited Talks & Presentations

Fair Scheduling and Resource Allocation for Public Services *Apr 2025*
USC Center for AI in Society, Seminar

Effect of Resources on the Efficiency-Fairness Tradeoff for Allocation Problems *Oct 2024*
ACM EAAMO 2024 Doctoral Consortium, Poster

Fair Decision-Making for Food Inspections *Apr 2024*
UIC Economics EARL Seminar

Fair Resource Allocation for Public Services *Oct 2023*
ACM EAAMO 2023 Doctoral Consortium, Poster

Experienced International TAs Panel *Aug 2022*
UIC New TA Orientation

Experienced International TAs Panel *Aug 2021*
UIC New TA Orientation

Cybersafety for Kids on Video Streaming Platforms *Jul 2019*
ACM Summer School on Cybersecurity and Analytics,
Indraprastha Institute of Information Technology Delhi

Teaching

Guest Lecturer, CS 401: Artificial Intelligence *Spring 2024*
University of Illinois Chicago

- Taught lectures on Machine Learning and Fairness in AI and ML to of 20+ graduate students.
- **Instructor:** Ian A. Kash

Graduate Teaching Assistant, CS 361: Computer Systems Programming *Fall 2019, Fall 2020, Spring 2021*
University of Illinois Chicago

- Designed, instructed, and evaluated programming labs and assignments for the Computer Systems Programming course, taken by 180+ undergraduate students in both in-class and remote environments.
- Received an average 4.48/5.0 in the evaluation feedback.
- **Instructor:** Chris Kanich

Lab Instructor, Summer School on Social Computing *Summer 2018*
International Institute of Information Technology Hyderabad

- Taught lab sessions for 50+ summer school participants on data collection using APIs, data analysis and constructing network graphs.
- Made the tutorial code public at precog-iiitd/social-computing-18.
- **Instructor:** Ponnurangam Kumaraguru

Teaching Assistant, CSE 501: Designing Human Centered Systems *Spring 2016*
Indraprastha Institute of Information Technology Delhi

- One of the select few undergraduate teaching assistants for a class taken by 80+ undergraduate and graduate students.
- Assisted in defining the course structure, designed and evaluated assignments and course projects.
- **Instructor:** Ponnurangam Kumaraguru

Professional Service

AIES *2025*
Program Committee

ICML *2025*
External Reviewer

Evaluating Evaluations for Generative AI Workshop, NeurIPS *2024*
Program Committee

ACM EAAMO *2024*
Program Committee

Aspiring PI Workshop, NSF SaTC *2024*
Organizing Volunteer

AAAI *2024*
External Reviewer

ACM EAAMO *2023*
Program Committee, Socials Facilitator

Paper Presentation and CRAFT Sessions, ACM FAccT *2023*
Student Volunteer

Algorithmic Fairness through the Lens of Causality and Privacy (AFCP) Workshop, NeurIPS *2022*
External Reviewer

Paper Presentation and CRAFT Sessions, ACM FAccT *2022*
Student Volunteer

CSCW *2022*
External Reviewer

Algorithmic Fairness through the Lens of Causality and Robustness (AFCR) Workshop, NeurIPS *2021*
External Reviewer

Leadership Activities

Working Groups Co-Leader *2022–2025*
EAAMO Bridges

- Organized six working groups that bring together 200+ researchers from diverse backgrounds around the world. The working groups focus on issues related to Law & Policy for Algorithms, Civic Participation, Climate Conservation, Conversations with Practitioners, Urban Data Science, and Inequality.
- Recruited participants for the six working groups, matched them based on fit, and polled them at the end

of the semester to improve the organizing process.

Treasurer *2020–2023*
UIC CS Graduate Student Association

- Treasurer and organizer for social and professional events for Computer Science graduate students at UIC.
- Received an award for exceptional service.

Student Mentor *2016*
Indraprastha Institute of Information Technology Delhi

- Mentored 10+ freshman students for the academic and non-academic issues faced by them.
- Received an award for exceptional service.

Career Coordinator *2014*
Training & Placement Cell, Indraprastha Institute of Information Technology Delhi

- Organized and managed the campus recruitment interviews with 35+ company officials for 250+ students of the senior batch.

Volunteer Work & Community Outreach

Volunteer *2025*
CicLAvia

- Helped organize and gather feedback at an event that opens streets for biking to 50,000+ attendees.

River Ranger *2023–2024*
Urban Rivers

- 8+ hours of Kayak cleaning on the Chicago River.

Volunteer *2023*
Chicago Ornithological Society

- Beach cleaning and invasive plant removal to support native birds and plants at the Montrose Point Bird Sanctuary.

Food Repacking Volunteer *2014*
Greater Chicago Food Depository

- Helped repack and deliver fresh produce to marginalized communities at one of the largest food banks in Chicagoland..

Teaching Volunteer *2014*
Pratham Delhi

- Taught underprivileged children science and mathematics in understaffed Pratham education centers.